

# Modular Series DOI and Relationship Map

Canonical series index - [10.5281/zenodo.21420798](https://doi.org/10.5281/zenodo.21420798)

## Abstract

Finite Possibility Mechanics (FPM) is a finite-resource mechanics of directed alternatives, formulated on a finite periodic cubic lattice. Each site carries bounded energy, a directed 3x3 route-cost ledger, a normalized nine-channel complex carrier, and local auxiliary state. Ordinary per-tick action is redistributed through a nearest-neighbour reversible Markov kernel and recorded in a signed ledger, yielding exact finite-graph results for local and global conservation, nodewise and regional continuity, finite propagation, equilibrium transport, and finite work capacity. The modular series separates this programme into six independently citable layers: foundations; executable reference semantics; carrier and information dynamics; a deterministic exact-integer sandbox; phenomenological bridges to quantum, thermodynamic, gravitational, cosmological, and electromagnetic observables; and an empirical audit and falsification protocol. A public Python simulator and machine-readable output reproduce the computational checks associated with Papers 01, 02, and 04. The formal and computational results establish properties of the stated model; bridge correspondences, calibrations, and retrospective comparisons are hypotheses and audits rather than independent evidence of physical validity. The series therefore presents both a reproducible construction and explicit empirical failure conditions.

## DOI map

| Record           | Scope and persistent identifier                                                                                                                                                                                              |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PAPER 01</b>  | FPM Foundations - Finite substrate, route ledger, local transport, and exact closure.<br><a href="https://doi.org/10.5281/zenodo.21420508">10.5281/zenodo.21420508</a>                                                       |
| <b>PAPER 02</b>  | Executable FPM - Reference runtime, numerical contract, and reproducible audit.<br><a href="https://doi.org/10.5281/zenodo.21420643">10.5281/zenodo.21420643</a>                                                             |
| <b>PAPER 03</b>  | Finite Carrier and Information Dynamics - Coherence, consolidation, information accounting, and finite allocation.<br><a href="https://doi.org/10.5281/zenodo.21420648">10.5281/zenodo.21420648</a>                          |
| <b>PAPER 04</b>  | Tensorless Exact-Ledger Sandbox - Deterministic integer policies and the boundary with reference FPM.<br><a href="https://doi.org/10.5281/zenodo.21420650">10.5281/zenodo.21420650</a>                                       |
| <b>PAPER 05</b>  | Phenomenological Bridges - Declared correspondences with quantum, gravitational, and cosmological observables.<br><a href="https://doi.org/10.5281/zenodo.21420652">10.5281/zenodo.21420652</a>                              |
| <b>PAPER 06</b>  | Empirical Audit and Falsification - Evidence levels, reproduction standards, and decisive empirical tests.<br><a href="https://doi.org/10.5281/zenodo.21420655">10.5281/zenodo.21420655</a>                                  |
| <b>SIMULATOR</b> | FPM Reference Python Simulator and Audit Results - public source and machine-readable audit output; supplements Papers 01, 02, and 04 only.<br><a href="https://doi.org/10.5281/zenodo.21420735">10.5281/zenodo.21420735</a> |

## Relationship key

|                 |                                                                                                                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Is derived from | Papers 01-06 and the simulator/result package identify the unified monolith as their source: <a href="https://doi.org/10.5281/zenodo.21352386">10.5281/zenodo.21352386</a> .                                  |
| Is part of      | Each modular paper and the simulator/result package belongs to this series index: <a href="https://doi.org/10.5281/zenodo.21420798">10.5281/zenodo.21420798</a> .                                             |
| Supplements     | The simulator/result package <a href="https://doi.org/10.5281/zenodo.21420735">10.5281/zenodo.21420735</a> supplements Papers 01, 02, and 04. No simulator relationship is asserted for Papers 03, 05, or 06. |

Reproducible source: <https://github.com/alxspiker/Finite-Possibility-Mechanics-Complete>